CHAPTER 5

Effective Annual Rate (EAR) & Yield Equivalence

Standardizing Compounding Frequencies, Nominal vs Effective Spreads, and Analytical Rate Inversion

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5.1 Definition & The Need for Standardized Yields

In global financial markets, interest rates are routinely quoted as an **Annual Percentage Rate (APR)** or nominal rate, but paired with disparate compounding frequencies (e.g., credit cards compounded monthly, corporate bonds semi-annually, money markets daily). Comparing nominal rates directly leads to severe financial mispricing. The **Effective Annual Rate (EAR)** standardizes all borrowing and investing yields to an annualized equivalent basis.

Definition 5.1 (Effective Annual Rate Formulation)

Let nominal annual interest rate be $r_{nom} \geq 0$ compounded n times per year.
The **Effective Annual Rate (EAR)** is the true annual yield satisfying $(1 + EAR) = (1 + r_{nom}/n)^n$:

$$EAR = (1 + r_{nom} / n)^n - 1$$

$$EAR_{cont} = \lim_{n \to \infty} (1 + r_{nom} / n)^n - 1 = e^{r_{nom}} - 1$$

where $EAR \geq r_{nom}$ for all $n \geq 1$, with equality holding if and only if $n = 1$ or $r_{nom} = 0$.

5.2 Rate Inversion & Yield Equivalence Across Frequencies

Analytical Rate Inversion: Given a desired effective annual yield EAR, what nominal rate r_{nom} must a financial product quote under compounding frequency n?
Inverting Definition 5.1 algebraically:

- $(1 + r_{nom} / n)^n = 1 + EAR \implies 1 + r_{nom} / n = (1 + EAR)^{1/n}$
- **Discrete Inversion:** $r_{nom}(n) = n\,[(1 + EAR)^{1/n} - 1]$
- **Continuous Inversion:** $e^{r_{cont}} = 1 + EAR \implies r_{cont} = \ln(1 + EAR)$.

Theorem 5.1 (Binomial Expansion of the Compounding Spread)

The spread between the Effective Annual Rate and the nominal rate $\Delta(n) = EAR - r_{nom}$ expands as compounding frequency increases, governed by the binomial second-order term:
$\mathbf{\Delta(n) \approx [(n - 1) / (2n)] \cdot r_{nom}^2}$

Proof: Expanding $(1 + r/n)^n$ via the Binomial Theorem: $1 + n(r/n) + [n(n–1)/2](r/n)^2 + \ldots = 1 + r + [(n–1)/(2n)]r^2 + O(r^3)$. Subtracting 1 and r yields the result. As $n \to \infty$, the maximum continuous spread converges to $(1/2)r^2$. ■

5.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: evaluating predatory consumer credit APR distortion, and constructing equivalent nominal yields across diverse compounding frequencies.

Example 5.1: Consumer Credit Card APR vs. NBFC Personal Loan

Problem Statement: A borrower compares two competing credit facilities:

- **Facility A (Credit Card):** Advertised nominal rate ***R* = 36.00% APR** compounded monthly ($n = 12$)
- **Facility B (NBFC Loan):** Advertised nominal rate ***R* = 39.50% APR** compounded annually ($n = 1$)

Which borrowing facility imposes the higher true economic borrowing cost?

Step-by-Step Analytical Solution:

Facility A (Credit Card):

- Nominal rate decimal: $r_{nom} = 0.36$, frequency $n = 12$.
- Periodic monthly rate: $i = 0.36 / 12 = 0.0300$ (3.00% per month).
- Applying the EAR formula:
  $EAR_A = (1 + 0.03)^{12} - 1 = (1.03)^{12} - 1$
  Since $(1.03)^{12} \approx 1.42576089$:
  $EAR_A = 1.42576089 - 1 = 0.42576089$ &implies; **42.58% Effective Annual Rate**.

Facility B (NBFC Loan):

- Since $n = 1$: $EAR_B = (1 + 0.395)^1 - 1 =$ **39.50% Effective Annual Rate**.

Economic Analysis & Decision:

- Cost discrepancy: $EAR_A - EAR_B = 42.58\% - 39.50\% =$ **+3.08% excess cost**.
- *Takeaway:* Despite Facility A marketing an APR that appears 3.50% lower than Facility B (36% vs 39.5%), its true annualized cost is over 3.0% higher due to monthly compounding velocity. This highlights why financial regulations require disclosure of Effective Annual Rates.

Example 5.2: Yield Equivalence Calibration Across Frequencies

Problem Statement: An institutional wealth fund sets an absolute performance hurdle of **EAR = 10.00%** effective annual yield. Determine the equivalent nominal rates r_{nom} that must be contracted under:

- **Case A:** Semi-annual compounding ($n = 2$)
- **Case B:** Monthly compounding ($n = 12$)
- **Case C:** Continuous compounding ($n \to \infty$)

Step-by-Step Analytical Solution:

Case A (Semi-Annual Compounding, $n = 2$):

- $r_{nom}(2) = 2 \times [(1 + 0.10)^{1/2} - 1]$
- $(1.10)^{0.5} \approx 1.04880885$
- $r_{nom}(2) = 2 \times (0.04880885) = 0.0976177$ &implies; **9.7618% per annum**.

Case B (Monthly Compounding, $n = 12$):

- $r_{nom}(12) = 12 \times [(1 + 0.10)^{1/12} - 1]$
- $(1.10)^{1/12} \approx 1.00797414$
- $r_{nom}(12) = 12 \times (0.00797414) = 0.0956897$ &implies; **9.5690% per annum**.

Case C (Continuous Compounding, $n \to \infty$):

- $r_{cont} = \ln(1 + EAR) = \ln(1.10)$
- $r_{cont} = 0.09531018$ &implies; **9.5310% per annum**.

Summary of Equivalence:

- 10.00% annual ≡ 9.7618% semi-annual ≡ 9.5690% monthly ≡ 9.5310% continuous.
- *Analytical Insight:* As compounding frequency increases, a lower nominal quoted rate suffices to achieve the exact same terminal dollar wealth.

5.4 Algorithmic Implementation in Python

Below is the complete implementation from models/time_value.py, providing bidirectional conversion between nominal rates and effective yields:

Listing: models/time_value.py — Yield Equivalence Engine

```
import math

def effective_annual_rate(nominal_rate: float, compounding_frequency: int) -> float:
    """
    Computes discrete EAR: EAR = (1 + r / n) ** n - 1.
    """
    if nominal_rate < 0.0 or compounding_frequency <= 0:
        raise ValueError("Nominal rate must be non-negative and frequency n >= 1.")
    periodic_rate = nominal_rate / compounding_frequency
    return float(((1.0 + periodic_rate) ** compounding_frequency) - 1.0)


def continuous_effective_annual_rate(nominal_rate: float) -> float:
    """Computes continuous EAR: EAR_cont = exp(r) - 1."""
    if nominal_rate < 0.0:
        raise ValueError("Nominal rate must be non-negative.")
    return float(math.exp(nominal_rate) - 1.0)


def nominal_rate_from_ear(ear: float, compounding_frequency: int) -> float:
    """Inverts EAR to nominal rate: r = n * [(1 + EAR) ** (1/n) - 1]."""
    if ear < 0.0 or compounding_frequency <= 0:
        raise ValueError("EAR must be non-negative and n >= 1.")
    return float(compounding_frequency * (((1.0 + ear) ** (1.0 / compounding_frequency)) - 1.0))
```

5.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Discrete Formula	$EAR = (1 + r/n)^n - 1$	Standardizes periodic yields to annual equivalent
Continuous Bound	$\lim_{n\to\infty} EAR = e^r - 1$	Asymptotic maximum effective return for given nominal rate
Monotonicity in *n*	$\partial(EAR)/\partial n > 0 \quad \forall\, r > 0$	Higher compounding frequency strictly expands yield spread
Inversion Operator	$r_{nom} = n[(1+EAR)^{1/n} - 1]$	Calibrates contract rates to target hurdle returns
Continuous Inversion	$r_{cont} = \ln(1 + EAR)$	Standard log-yield transformation in financial modeling